Macroeconomics

Time: 120 minutes

Total: 10 points

Instructor: Quang-Thanh Tran

Main Exam (Version 1)

This exam contains two parts.

Using AI is strictly prohibited. Reviewing past lecture notes are possible.

---

1. (4 points) Consider the following Ramsey-style dynamic optimization problem

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

where

$$u(c_t) = -\frac{1}{\eta} e^{-\eta c_t}$$

subject to

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t, \qquad c_t > 0.$$

given $\beta, \eta, \delta \in (0,1)$.

Answer the following questions

(a) Write the Lagrangian and derive the Euler equation of the optimal consumption-saving.

(b) Compute the intertemporal elasticity of substitution $IES = -\dfrac{u'(c)}{c\,u''(c)}$. What happens to the IES as $c \to 0$ and $c \to \infty$? Explain the economic meaning.

(c) Assuming an interior solution ($c_t > 0$), derive the change in consumption $c_{t+1} - c_t$. What does a higher η imply for the consumption smoothing behavior[1]? Derive the steady state level of consumption c^*. Assume that $f(k) = Ak^\alpha$. What is the steady state level of capital k^*? Does it depend on parameter η?

(d) Assume that the government makes a one-time positive transfer to households at $t = 0$. What would be the likely response of poor household (low c) and rich household (high c) regarding consumption now (and saving for later)?

[1]Note: more consumption smoothing means households tend to dislike sudden change in consumption and prefer to maintaining stable consumption over the lifetime.

2. (6 points) We want to study the availability and effectiveness of contraceptives (các biện pháp phòng tránh thai không mong muốn) on fertility and the macroeconomy.

Demography

Time is discrete, $t = 0, 1, 2, \ldots$ In each period N_t identical young agents are born; each lives for two periods—young (period t) and old (period $t+1$). The population of young agents evolves according to

$$N_{t+1} = n_t N_t, \tag{1}$$

where n_t is the *realized* fertility rate of a young household in period t.

Fertility

Realized births consist of two components:

$$n_t = n_t^d + (1 - \kappa)u, \tag{2}$$

where

- $n_t^d \geq 0$ is *desired* (or intended) fertility, chosen by the household;
- $u > 0$ is the *unintended fertility shifter*, a biological parameter capturing the maximum number of unintended births in the absence of contraception;
- $\kappa \in [0, 1]$ is *contraceptive effectiveness*: the fraction of unintended births that contraception prevents.

When $\kappa = 0$, no contraception is used and realized fertility is $n_t = n_t^d + u$. When $\kappa = 1$, contraception is perfect, i.e., the household can perfectly control births and $n_t = n_t^d$. When $\kappa < 1$, the contraceptive is not perfect, resulting in realized births larger than intended.

Preferences

An important assumption is that households derive utility only from *desired* children, while the budget constraint is determined by *realized* births.

Each young household maximizes lifetime utility

$$U_t = \ln c_t^y + \beta \ln c_{t+1}^o + \gamma \ln n_t^d, \tag{3}$$

where $\beta, \gamma \in (0, 1)$ and:

- c_t^y is young-age consumption,
- c_{t+1}^o is old-age consumption,
- $\beta > 0$ is the subjective discount factor,
- $\gamma > 0$ is the weight on desired children in utility.

Production

A representative firm produces output Y_t using capital K_t and labor L_t with a Cobb–Douglas technology:

$$Y_t = A K_t^{\alpha} L_t^{1-\alpha}, \qquad \alpha \in (0, 1), \;\; A > 0. \tag{4}$$

Capital depreciates fully within one period ($\delta = 1$).

Each young worker supplies one unit of labor inelastically, so $L_t = N_t$. For simplicity, we are assuming that taking care of children do not takes time. The cost per child is a portion of adult's income.

Budget constraint

Raising children requires a goods cost proportional to the household's wage income. Each child (desired or unintended) costs a fraction $\phi \in (0,1)$ of the wage w_t. Thus the total child-rearing expenditure is $\phi n_t w_t$, and the household's budget constraint is

$$c_t^y + s_t = (1 - \phi n_t) w_t, \tag{5}$$

$$c_{t+1}^o = R_{t+1} s_t, \tag{6}$$

where $s_t \geq 0$ denotes saving and R_{t+1} is the gross return on capital in $t+1$ (the interest factor).

Answer the following questions

(a) Write the firm's profit maximization problem and derive the optimal capital rental rate R_t and wage rate w_t.

(b) Solve the household problem that maximizes lifetime utility by choosing the optimal c_t^y, s_t, n_t^d. You should be able to derive c_t^y, s_t, n_t^d as functions of w_t. What is the total fertility n_t?

(c) Is it possible for the desired fertility to fall to zero? Find the condition for $n_t^d > 0$ always.

(d) The capital market clears according to

$$K_{t+1} = N_t s_t$$

Write the law of motion for capital per worker k_{t+1}.

(e) What are the steady-state values of capital per worker, desired and realized fertility?

(f) What happens to the steady-state values of capital per worker, desired fertility, and realized fertility when there is an increase in κ (i.e., improved contraceptive effectiveness)? Can you give an intuitive explanation of such changes?

END OF EXAM
